Customer Shopping Behavior Analysis

Data Analytics Project using Python, SQL & Power BI

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1. Project Overview

Understanding customer purchasing behavior is essential for businesses to improve marketing strategies, increase revenue, and enhance customer retention.

This project analyzes customer shopping data to uncover insights about purchasing patterns, customer loyalty, product performance, and subscription behavior.

The analysis was conducted using **Python for data exploration**, **SQL for business query analysis**, and **Power BI for dashboard visualization**. The goal of this project is to transform raw customer transaction data into meaningful insights that can support data-driven business decisions.

2. Business Problem

Retail businesses generate large volumes of customer transaction data, but extracting actionable insights from this data remains a challenge.

This project aims to answer key business questions such as:

- What factors influence customer purchase behavior?
- Which customer segments generate the most revenue?
- Do repeat customers subscribe more frequently?
- How do discounts impact purchasing behavior?
- Which products and categories perform best?

By answering these questions, businesses can optimize their marketing strategies and improve customer engagement.

3. Dataset Description

The dataset used in this project contains customer shopping transaction records including demographic information, purchase behavior, and product details.

Key Features:

- Customer demographics (Age, Gender, Location, Subscription Status)
- Product details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Purchase behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency Of Purchases, Review Rating, Shipping Type)
- Missing data: 37 values in Review Rating Column

Dataset Summary

- **Total Records:** 3900
- **Total Column:** 18
- **Data Type:** Structured transactional dataset

4. Tools & Technologies Used

The following tools were used during the analysis process:

Programming & Analysis

- **Python**
  - Pandas
  - NumPy
  - Matplotlib
  - Seaborn

Database Querying

- **SQL**
  - Data aggregation
  - Customer segmentation
  - Business queries

Data Visualization

- **Power BI**
  - Interactive dashboards
  - KPI tracking
  - Business insights visualization

5. Data Cleaning & Preparation

Before performing the analysis, several data preprocessing steps were carried out to ensure the dataset was clean and analysis-ready.

Data Preparation Steps:

- Loading data and initial exploration
- Checked for missing values
- Removed duplicate records
- Standardized column formats
- Converted data types where necessary
- Created new analytical features

Two important features created during the analysis were **Customer Segment**, which classifies customers based on their previous purchase behavior and **Age Group,** to improve demographic analysis and business interpretability, we segment customers into age groups. Age-based segmentation allows the business to tailor marketing strategies and product offerings to different life stages.

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used df.info() to check structure and df.describe() for summary statistics.

[2]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900.000000	3900	3900
2	NaN	6	7
No	NaN	PayPal	Every 3 Months
2223	NaN	677	584
NaN	25.351538	NaN	NaN
NaN	14.447125	NaN	NaN
NaN	1.000000	NaN	NaN
NaN	13.000000	NaN	NaN
NaN	25.000000	NaN	NaN
NaN	38.000000	NaN	NaN
NaN	50.000000	NaN	NaN

- **Handling Missing/null values:** Checked for missing values and imputed missing values in the Review Rating column using the median rating within each product category to reduce bias from extreme values.
- **Column Standardization:** Renamed columns to **snake_case** for better readability and documentation.
- **Customer segment creation:**

```
[33]: conditions = [
          df['previous_purchases'] == 1,
          df['previous_purchases'].between(2, 10),
          df['previous_purchases'] > 10
      ]

      choices = ['New', 'Returning', 'Loyal']

      df['customer_segment'] = np.select(
          conditions,
          choices,
          default='Unknown'
      )

      customer_segment = (
          df.groupby('customer_segment')['customer_id']
          .count()
      )
      customer_segment
```

```
[33]: customer_segment
      Loyal        3116
      New            83
      Returning     701
      Name: customer_id, dtype: int64
```

- **Age Group Creation:**

The following age groups are defined based on common retail demographic segmentation:

- Young Adult: 18–25
- Adult: 26–40
- Middle Aged: 41–60
- Senior: 61 and above

```
[12]: bins = [0, 25, 40, 60, 100]
      labels = ['Young Adult', 'Adult', 'Middle Aged', 'Senior']

      df['age_group'] = pd.cut(df['age'], bins=bins, labels=labels)
```

- **Database Integration:** Connected python script to MySQL and loaded the cleaned DataFrame Into database for SQL analysis.

6. Exploratory Data Analysis (Python)

Exploratory data analysis was conducted using Python to identify patterns and relationships in customer purchasing behavior.

Several visualizations were created to understand spending trends, customer demographics, and purchase distributions.

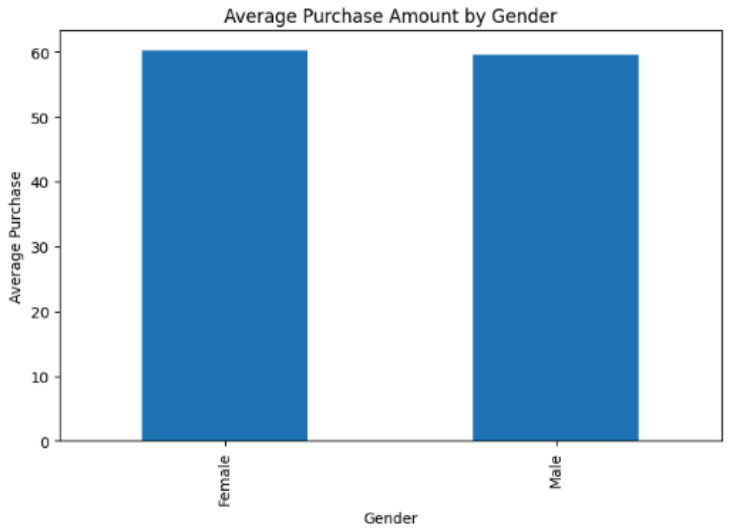
Key Observations:

- Female customers showed slightly higher average purchase value than male customers.
- Loyal customers represented the largest portion of the customer base.
- Adult and Middle Aged customers contribute the highest share of total revenue.

[61]:
```
avg_purchase_by_gender = (
    df.groupby('gender')['purchase_amount']
    .mean()
)
avg_purchase_by_gender
```

[61]:
```
gender
Female    60.249199
Male      59.536199
Name: purchase_amount, dtype: float64
```

[62]:
```
plt.figure(figsize=(8, 5))
avg_purchase_by_gender.plot(kind='bar')
plt.title('Average Purchase Amount by Gender')
plt.xlabel('Gender')
plt.ylabel('Average Purchase')
plt.show()
```

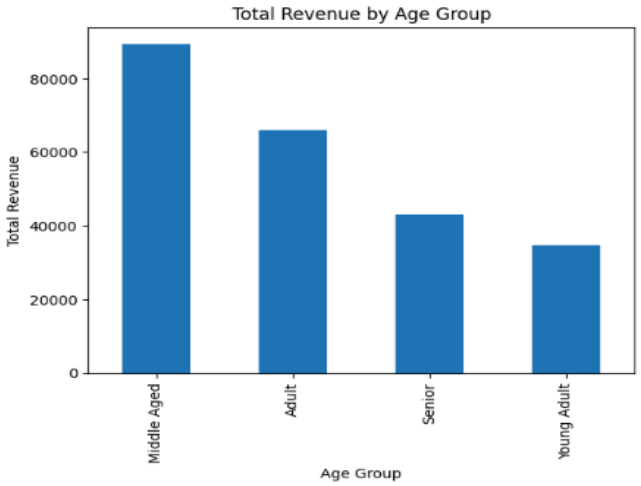


[55]:
```
age_group_revenue = (
    df.groupby('age_group')['purchase_amount']
    .sum()
    .sort_values(ascending=False)
)
age_group_revenue
```

```
C:\Users\mohda\AppData\Local\Temp\ipykernel_3700\2121822882.py:2: FutureWarning: The default of observed=False is deprecated and will be changed t
rue in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this wa
ng.
  df.groupby('age_group')['purchase_amount']
```

[55]:
```
age_group
Middle Aged    89445
Adult          65842
Senior         43164
Young Adult    34630
Name: purchase_amount, dtype: int64
```

[56]:
```
plt.figure()
age_group_revenue.plot(kind='bar')
plt.title('Total Revenue by Age Group')
plt.xlabel('Age Group')
plt.ylabel('Total Revenue')
plt.show()
```



7. Data Analysis using SQL

SQL was used to answer several business questions and validate insights derived from the Python analysis.

1. **Revenue by gender:** Compared total revenue generated by male vs. female customers.

	gender	total_revenue
0	Male	157890.0
1	Female	75191.0

2. **High spending discount users:** Identified customers who used discounts but still spent more than the average purchase amount.

	customer_id	total_spent
0	2	64.0
1	3	73.0
2	4	90.0
3	7	85.0
4	9	97.0
...	...	...
834	1667	64.0
835	1671	73.0
836	1673	73.0
837	1674	62.0
838	1676	90.0

839 rows × 2 columns

3. **Top 5 products by rating:** Found products with the highest average review ratings.

	item_purchased	avg_rating
0	Gloves	3.86
1	Sandals	3.84
2	Boots	3.82
3	Hat	3.80
4	Handbag	3.78

4. **Shipping type comparision:** Compared average purchase value between Standard and Express shipping.

	shipping_type	avg_purchase_value
0	Express	60.48
1	Standard	58.46

5. **Subscribers vs. Non-subscribers:** Compared average spend and total revenue between subscribers and non-subscribers.

	subscription_status	total_revenue	avg_spend
0	No	170436.0	59.87
1	Yes	62645.0	59.49

6. **Customer segmentation:** Segment customers into New, Returning and Loyal based on their purchase history.

	customer_segment	total_customer
0	Loyal	3116
1	New	83
2	Returning	701

7. **Top 3 products per category:** Listed most purchased products of each category.

	item_purchased	category	total_orders	ranked_item
0	Jewelry	Accessories	171	1
1	Sunglasses	Accessories	161	2
2	Belt	Accessories	161	3
3	Blouse	Clothing	171	1
4	Pants	Clothing	171	2
5	Shirt	Clothing	169	3
6	Sandals	Footwear	160	1
7	Shoes	Footwear	150	2
8	Sneakers	Footwear	145	3
9	Jacket	Outerwear	163	1
10	Coat	Outerwear	161	2

8. **Repeat Buyers & Subscriptions:** Checked customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

	repeat_buyers	subscription_status
0	2518	No
1	958	Yes

9. **Revenue by Age Group:** Calculated total revenue contribution of each age group.

	age_group	total_revenue
0	Middle Aged	89445.0
1	Adult	65842.0
2	Senior	43164.0
3	Young Adult	34630.0

10. **Revenue by Category:** Checked most revenue contribution product categories to overall revenue.

	category	total_revenue
0	Clothing	104264.0
1	Accessories	74200.0
2	Footwear	36093.0
3	Outerwear	18524.0

8. Power BI Dashboard

An interactive Power BI dashboard was developed to present insights in a visual and easy-to-understand format.

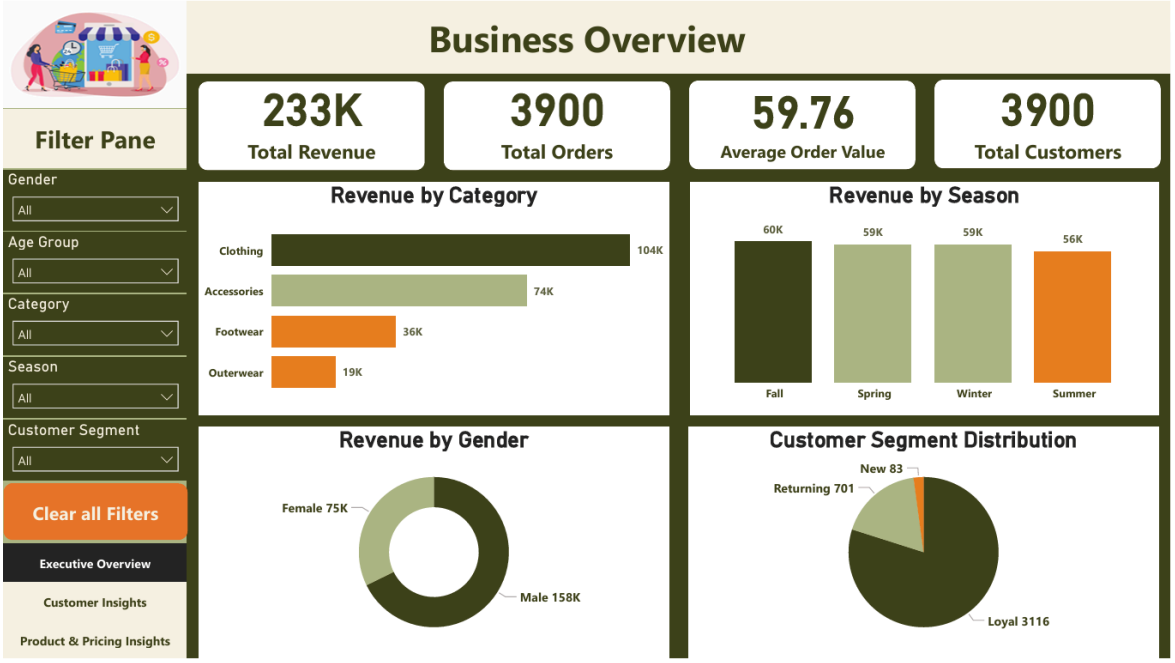
The dashboard is divided into **three main pages**.

Page 1 — Business Overview

This page provides a high-level overview of business performance.

Key metrics displayed include:

- Total Revenue
- Total Orders
- Average Order Value
- Total Customers

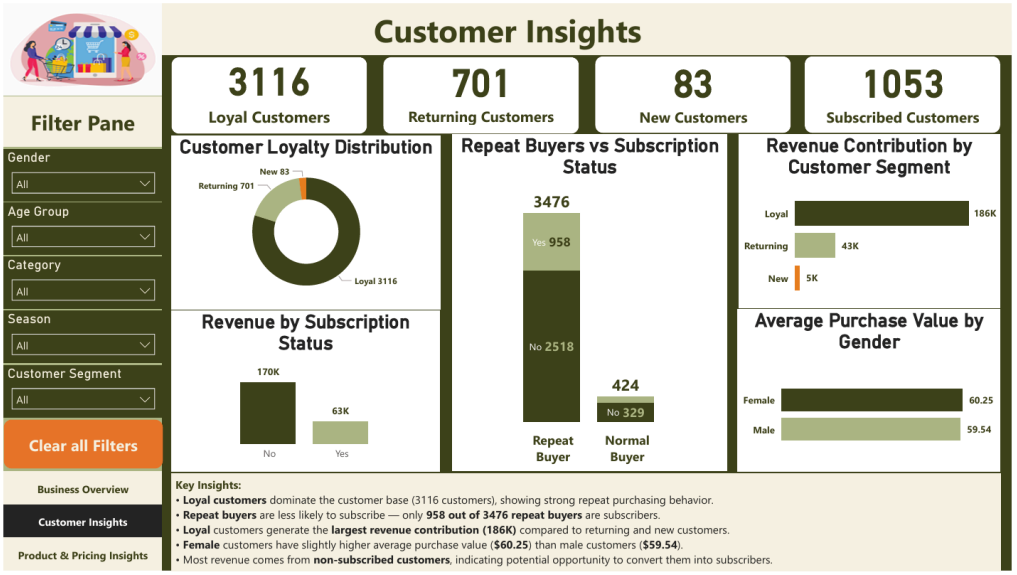


Page 2 — Customer Insights

This page focuses on understanding customer demographics and purchasing behavior.

Key analyses include:

- Gender spending comparison
- Loyalty distribution
- Customer segments
- Subscription behavior

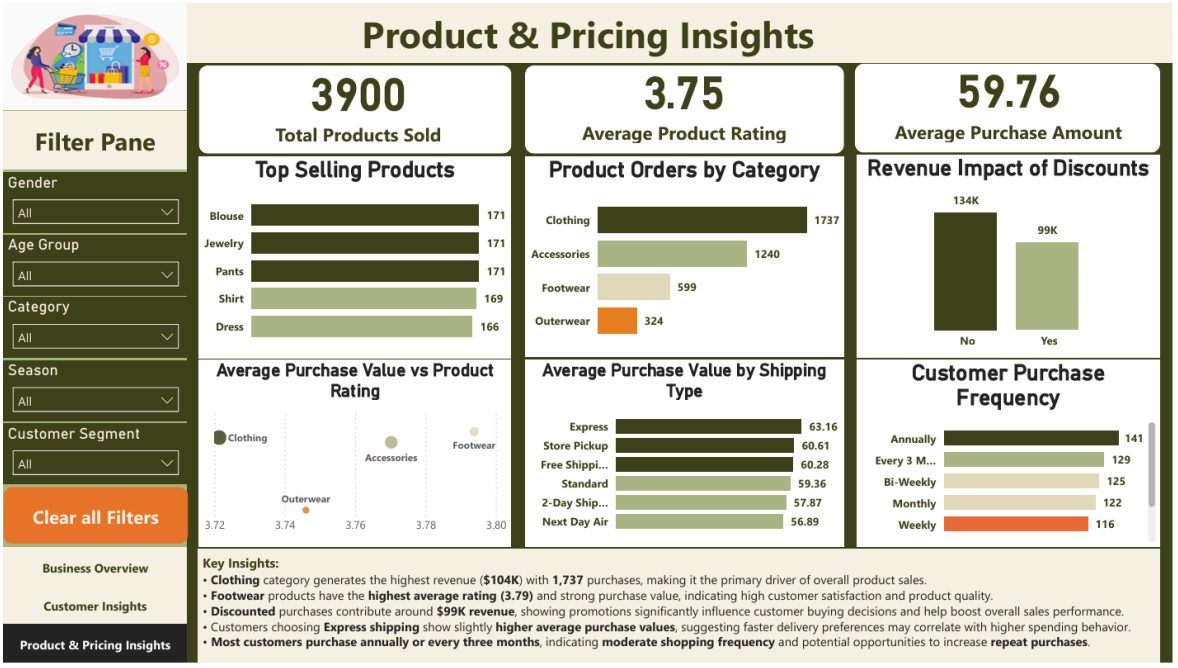


Page 3 — Product & Pricing Insights

This page analyzes product performance and pricing impact.

Key analyses include:

- Top performing products
- Discount impact on purchase value
- Product rating influence
- Shipping type analysis
- Purchase frequency patterns



9. Key Insights

Based on the analysis, several important insights were identified.

- Loyal customers make up the largest segment of the customer base.
- Female customers have slightly higher average purchase values than male customers.
- Discounts do not significantly increase purchase value.
- Customers who frequently purchase are more likely to subscribe.
- Products with higher ratings tend to generate higher revenue.

These insights highlight the importance of customer retention and product quality in driving revenue growth.

10. Business Recommendations

Based on the findings, the following business strategies are recommended.

1. Focus on Customer Retention

Since loyal customers represent the largest segment, businesses should implement loyalty programs and personalized marketing campaigns.

2. Target Repeat Buyers for Subscriptions

Repeat customers show higher subscription potential. Businesses can offer targeted subscription promotions to increase recurring revenue.

3. Optimize Discount Strategy

Since discounts do not significantly increase purchase value, businesses should focus on strategic promotions instead of heavy discounting.

4. Promote Highly Rated Products

Products with higher ratings tend to perform better. Highlighting these products in marketing campaigns can boost sales.

11. Conclusion

This project demonstrates how data analytics can be used to uncover valuable insights into customer shopping behavior.
By combining **Python for data analysis**, **SQL for business queries**, and **Power BI for visualization**, the project successfully transformed raw customer data into actionable business intelligence.
The insights derived from this analysis can help businesses improve customer engagement, optimize pricing strategies, and enhance overall revenue performance.